

Caitlin Chiang

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Education

Singapore University of Technology and Design (SUTD)

Singapore

Bachelor of Science in Design and Artificial Intelligence

Sept 2021 - May 2025

Minors in Computer Science and Digital Humanities

- Design in Artificial Intelligence Pillar Batch Representative
- Experienced in software engineering, artificial intelligence, client-partnered projects, and UX design.

Work Experiences

SOFTWARE ENGINEER, Xweave.io | Singapore

May 2025 - Present

Fintech building blockchain-powered infrastructure for cross-border payments.

- Architected and delivered secure integrations with multiple digital asset exchange providers, enabling scalable cross-border payment orchestration through standardized REST APIs.
- Designed and productionized an event-driven webhook architecture to provide reliable, real-time transaction synchronization across distributed payment systems.
- Architected and implemented a modular mid-market rate service using object-oriented design principles, enabling efficient retrieval, persistence, and synchronization of fiat and cryptocurrency exchange rates.
- Led and mentored two software engineers through technical guidance, code reviews, and architectural discussions, improving engineering quality and delivery velocity.

AI SOFTWARE ENGINEER, Liminal | Singapore

Jan 2025 - April 2025

Temasek-backed venture capital.

- Engineered a production-ready full-stack multi-agent AI platform enabling configurable workflows, dynamic query execution, and customizable result management.
- Developed a responsive React-based chat interface featuring editable tables, configurable schemas, and real-time data visualization to improve user productivity.
- Architected a modular backend platform integrating multiple LLM providers, prompt engineering strategies, and extensible AI workflows for scalable feature development.

SOFTWARE ENGINEER, Moya | Singapore

Oct 2024 - Mar 2025

Climate technology startup developing data-driven solutions for emissions reduction and energy research.

- Engineered automated data processing pipelines to aggregate, transform, and consolidate large CSV datasets into standardized analytical outputs and master datasets.
- Designed reusable scripts to generate abatement curves and analytical datasets, improving data processing efficiency and consistency.

SOFTWARE ENGINEER, Jublia | Singapore

Aug 2023 - Dec 2023

Event technology company providing platforms for event management, networking, and attendee engagement.

- Redesigned a user-centric dashboard experience, improving usability and application navigation.
- Modernized a legacy Vue.js application by refactoring reusable components into a modular, maintainable architecture that improved scalability.
- Automated database integrity validation through scheduled monitoring scripts, improving operational reliability and reducing manual verification efforts.

SOFTWARE ENGINEER, SUTD campusX | Singapore

Jan 2023 - Dec 2023

University research initiative developing a human-centric cyber-physical learning environment.

- Collaborated with university faculty and cross-functional stakeholders to architect and deliver an AI-powered learning platform enhancing educational workflows and student engagement.
- Engineered a modular application framework configurable for multiple university courses.
- Defined the application's system architecture and relational database design, emphasizing scalability, maintainability, and long-term extensibility.

SOFTWARE ENGINEER, Dashlabs.ai | Philippines

Dec 2020 - Dec 2021

Y-Combinator healthtech startup building clinical analytics and healthcare software solutions.

- Led and mentored a team of seven software engineers, providing technical leadership, architectural guidance, and code reviews for healthcare analytics solutions.
- Collaborated with medical professionals and cross-functional stakeholders to translate complex clinical requirements into scalable analytical software.
- Delivered over 30 production features that streamlined healthcare workflows and improved operational efficiency for medical professionals.
- Engineered over 20 interactive analytics dashboards and architected data aggregation pipelines that transformed complex healthcare datasets into actionable insights and meaningful visualizations for medical professionals.

SOFTWARE ENGINEER, Philippine Red Cross | Philippines

Jul 2021 - Oct 2021

Humanitarian organization providing emergency, healthcare, and disaster response services.

- Developed automated module-level testing scripts to improve software reliability, maintainability, and deployment confidence.

FREELANCE SOFTWARE ENGINEER, Dragonpay Corporation | Philippines

Jun 2021 - Jul 2021

Leading Philippine payment gateway enabling digital payments across banks and payment channels.

- Developed and deployed an internal employee wellness platform, streamlining organizational health tracking and reporting.
- Engineered an internal expense management application that automated financial workflows and improved operational efficiency for the finance team.

FREELANCE SOFTWARE ENGINEER | Philippines

Mar 2020 - Jun 2021

Clients: The Rolls Kitchen, The Busy Bee, Soren Apparel

- Partnered directly with business owners and executives to translate business objectives into scalable web application solutions.
- Designed and developed end-to-end e-commerce platforms spanning customer-facing experiences and administrative management systems.
- Architected application databases and backend systems with a focus on maintainability, scalability, and long-term growth.

Academic Client Partnerships

Urban Redevelopment Authority (URA)

Sept 2024 - Dec 2024

AI-Powered Public Engagement Platform | Team Lead and Software Engineer

- Led a team of four engineers to architect and deliver an AI-powered public engagement platform that transformed user ideas into AI-generated visual concepts.
- Integrated advanced generative AI techniques, including prompt engineering and image inpainting, to improve image quality and user experience.
- Designed a configurable, modular application architecture enabling administrators to customize engagement workflows without code changes.
- Translated UX designs into production-ready software while ensuring consistency between design specifications and implementation.

Development Bank of Singapore Limited (DBS)

Jan 2024 - May 2024

Queue Simulation & Operational Planning Dashboard | Team Lead and Software Engineer

- Led a team of five engineers to develop a configurable web-based queue simulation platform supporting operational planning across banking branches.
- Architected a scalable and modular system design that enabled flexible simulation scenarios and future platform extensibility.

Procter & Gamble (P&G)

May 2023 - Aug 2023

Consumer Analytics & Data Visualization Platform | Technical Lead and Software Engineer

- Led a team of six engineers to deliver a data visualization platform supporting consumer behavior analysis for P&G's R&D organization.
- Engineered automated CSV ingestion and processing pipelines that transformed raw datasets into interactive analytical visualizations.
- Developed interactive dashboards enabling researchers to compare multi-dimensional datasets through configurable timelines, charts, and visual analytics.
- Designed consumer journey visualization features that uncovered behavioral trends and supported data-driven product decisions.
- Integrated AI-powered image analysis techniques to extract visual attributes and emotional signals from consumer-submitted images, enabling deeper product evaluation and research insights.

Additional Information

Technical Skills

- Languages: JavaScript, TypeScript, Python, Ruby, HTML, CSS
- Frontend: React.js, Next.js, Vue.js, Material UI, Chakra UI
- Backend: Node.js, Express.js, Flask, Ruby on Rails
- AI/LLM: LangChain, LangGraph
- Databases: PostgreSQL, MySQL, MongoDB, Firebase
- APIs: REST APIs, GraphQL (Apollo)
- Cloud: Amazon Web Services (AWS), Google Cloud Platform (GCP), Cloudinary, Netlify
- Developer Tools: Git, Cypress, Snakemake